

Physiologic brain activity causes DNA double-strand breaks in neurons, with exacerbation by amyloid- β

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We show that a natural behavior, exploration of a novel environment, causes DNA double-strand breaks (DSBs) in neurons of young adult wild-type mice. DSBs occurred in multiple brain regions, were most abundant in the dentate gyrus, which is involved in learning and memory, and were repaired within 24 h. Increasing neuronal activity by sensory or optogenetic stimulation increased neuronal DSBs in relevant but not irrelevant networks. Mice transgenic for human amyloid precursor protein (hAPP), which simulate key aspects of Alzheimer's disease, had increased neuronal DSBs at baseline and more severe and prolonged DSBs after exploration. Interventions that suppress aberrant neuronal activity and improve learning and memory in hAPP mice normalized their levels of DSBs. Blocking extrasynaptic NMDA-type glutamate receptors prevented amyloid- β (A β)-induced DSBs in neuronal cultures. Thus, transient increases in neuronal DSBs occur as a result of physiological brain activity, and A β exacerbates DNA damage, most likely by eliciting synaptic dysfunction.

Neuronal DNA damage may contribute to cognitive aging and the pathogenesis of neurodegenerative disorders such as Alzheimer's disease^{1,2}. Aging, the main risk factor for these disorders, is associated with a progressive increase in markers of DNA breaks in neurons³. One such marker is phosphorylation of the histone protein H2A variant X at Ser139 (resulting in γ H2A.X), focal accumulations of which reliably identify DNA DSBs^{4,5}.

Many studies of γ H2A.X have focused on the relationship between severe DNA damage and the activation of cell death pathways^{1,4–6}. However, γ H2A.X may also help recruit the DNA repair machinery and fulfill physiological functions in epigenetic processes that regulate chromatin structure and gene expression. For example, γ H2A.X appears to be involved in mitosis, meiosis, lymphocyte development, neural development and adult neurogenesis^{7–10}. In proliferating cells, its activation is typically associated with a pause in cell cycle progression⁴. In cultures of differentiated neurons, γ H2A.X increases transiently after stimulation with NMDA¹¹, suggesting a potential link between neuronal activity and DSBs. Phosphorylation of H2A.X has been identified as a predictor of neuronal death and an early marker of nonlethal neuronal harm caused by kainate-induced epileptic activity in rats¹².

Here we used γ H2A.X measurements and related approaches to examine the causes, extent and duration of neuronal DSBs in the brains of mice under physiological and pathophysiological circumstances. For physiological stimulation, we allowed mice to explore a novel environment, which causes widespread, transient neuronal activation^{13,14}. The natural behavior, visual stimulation in anesthetized mice and optogenetic activation of striatal neurons in awake, behaving mice each increased neuronal DSBs in specific brain regions

of wild-type mice. We also investigated DSBs in hAPP transgenic mice from line J20 (hAPP mice). These mice exhibit several features of Alzheimer's disease, including pathological elevation of A β peptides in the brain, age-dependent deficits in learning and memory, behavioral abnormalities, synaptic deficits, aberrant neuronal network activity, formation of neuritic amyloid plaques, and inflammatory reactions of astrocytes and microglia^{15–19}. Virtually all of these phenotypes have also been identified in other lines of hAPP transgenic mice^{18,20–22}. In comparison to wild-type controls, hAPP mice had more neurons with DSBs at baseline, as well as greater and more prolonged increases in DSBs after exploration of a novel environment. Suppression of aberrant network activity in hAPP mice prevented and reversed the baseline increase in neuronal DSBs. Consistent with these *in vivo* findings, A β oligomers caused DSBs to form in primary neuronal cultures, and this effect could be prevented by blocking extrasynaptic NMDA-type glutamate receptors (NMDARs). We conclude that transient increases in neuronal DSBs are an integral component of physiological brain activity and that A β exacerbates DNA damage by eliciting aberrant synaptic activity.

RESULTS

Pathological levels of A β increase neuronal γ H2A.X

To determine whether pathological elevation of A β causes neuronal DNA damage, we measured well established markers of DSBs in brain sections of hAPP mice and in primary neuronal cultures exposed to A β oligomers (Fig. 1). We counted cells that had at least one γ H2A.X-immunoreactive focus in the nucleus (Fig. 1a) in several brain regions of hAPP-J20 transgenic and wild-type mice at 6 months of age, when hAPP mice from this line have cognitive deficits and when

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Figure 1 A β increases neuronal γ H2A.X formation *in vivo* and *in vitro*. (a,b) γ H2A.X-positive foci in immunostained brain sections from 6-month-old wild-type (WT) and hAPP mice were visualized by confocal microscopy. (a) Images from the entorhinal cortex of an hAPP mouse. γ H2A.X-positive foci (green) were localized to nuclei labeled with DAPI (blue) and were found primarily in cells co-labeled with an antibody to the neuronal marker NeuN (red, right panel). Middle panel shows a higher magnification of the area in the white box in the left panel. Scale bars, 10 μ m. (b) Numbers of cells with γ H2A.X-positive foci per section (averaged from three sections per mouse) in different brain regions of WT and hAPP mice ($n = 8$ mice per genotype). (c) Western blot analysis of γ H2A.X in cultures of primary forebrain neurons from WT mice exposed to 1 μ M A β oligomers (+) or vehicle (–) for 4 h. α -Tubulin served as a loading control. Left, a representative western blot; right, quantification ($n = 8$ –12 wells per condition from four independent experiments). (d) Representative example of an γ H2A.X-positive focus (green) also labeled with anti-53BP1 (red) in a DAPI-labeled (blue) neuronal nucleus in the dentate gyrus of a 6-month-old hAPP mouse. Scale bar, 10 μ m. ** $P < 0.01$, *** $P < 0.001$ (two-tailed, unpaired Student's *t*-test). Error bars represent means $\pm$ s.e.m.

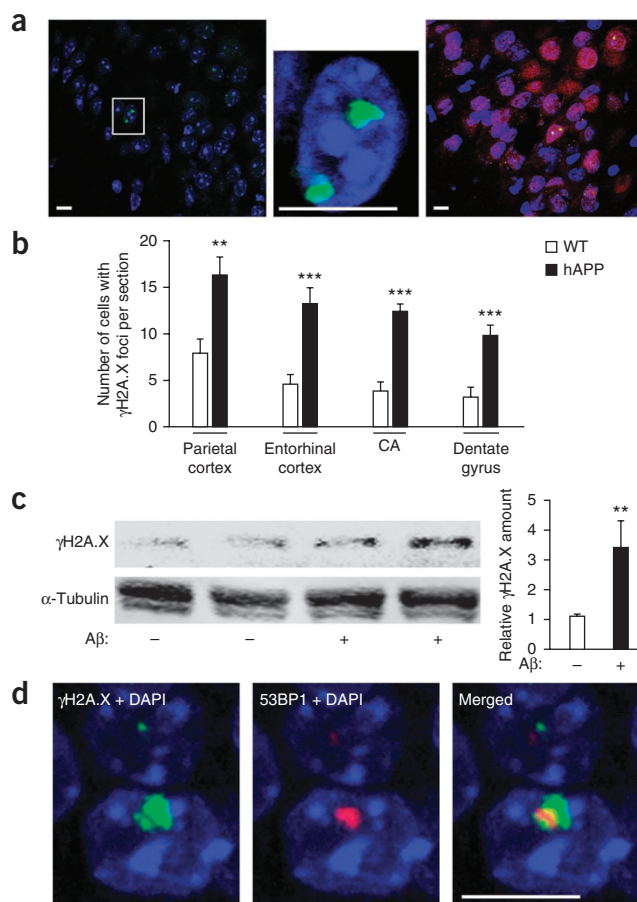
their brains contain pathogenic soluble A β assemblies but few amyloid deposits^{15,16}. In all brain regions studied, hAPP mice had 2.0–3.5 times more γ H2A.X-positive cells than wild-type controls (Fig. 1b). The γ H2A.X foci were located primarily in neurons, as shown by co-labeling of brain sections for the neuronal marker NeuN (Fig. 1a). Even at 1.0–1.5 months of age, before hAPP mice show behavioral abnormalities, the number of γ H2A.X-positive cells in the hippocampus and neocortex was higher in hAPP than wild-type mice (Supplementary Fig. 1), indicating that this alteration precedes—and thus might contribute to—cognitive decline.

In addition to elevated A β , hAPP mice also have elevation of other hAPP metabolites in the brain. We therefore determined whether A β oligomers were sufficient to increase neuronal γ H2A.X formation in cultured primary neurons from wild-type mice. Adding A β oligomers to the cultures increased γ H2A.X threefold (Fig. 1c).

Exploratory behavior increases neuronal γ H2A.X foci in mice

In the course of our experiments, we observed that exploration of a novel environment increased the number of neurons with γ H2A.X foci in mice (Fig. 2). This behavior is well known to increase neuronal activity in several brain regions, as reflected by widespread increases in neuronal expression of the activity-regulated cytoskeletal protein Arc^{13,14}. Unexpectedly, even wild-type mice that explored a novel environment had more γ H2A.X-positive neurons in various brain regions, particularly in the dentate gyrus, than controls that remained in their home cages (Fig. 2a,b and Supplementary Fig. 2a–c). Exploration also increased dentate Spo11 (Supplementary Fig. 2d,e), an endonuclease known to cause physiological DSBs during meiosis in the testis²³. In wild-type mice that explored the novel environment, the number of γ H2A.X-positive neurons returned to baseline after a 24-h recovery period in their home cage (Fig. 2a,b and Supplementary Fig. 2a–c).

As expected, hAPP mice had more γ H2A.X-positive neurons than wild-type mice under control (home cage) conditions (Fig. 2c and Supplementary Fig. 2a–c). Like wild-type controls, hAPP mice showed an increase in γ H2A.X-positive neurons after exploring the novel environment, particularly in the parietal cortex and dentate gyrus (Fig. 2c and Supplementary Fig. 2a–c). Student's *t*-test revealed no significant differences in the number of γ H2A.X-positive neurons between hAPP and wild-type mice after exploration (Fig. 2b,c). However, because of their increased DNA damage at baseline, hAPP mice tended to have smaller relative increases in



γ H2A.X-positive neurons after exploration than wild-type mice (Fig. 2b,c and Supplementary Fig. 2a–c). Like wild-type mice, hAPP mice had fewer γ H2A.X-positive neurons after a 24-h recovery than immediately after the novel environment exploration (Fig. 2b,c and Supplementary Fig. 2a–c). However, in the dentate gyrus and parietal cortex of hAPP mice, the decline was less complete than in wild-type mice and did not reach baseline (Fig. 2c and Supplementary Fig. 2a–c).

A β - and exploration-induced neuronal DSBs

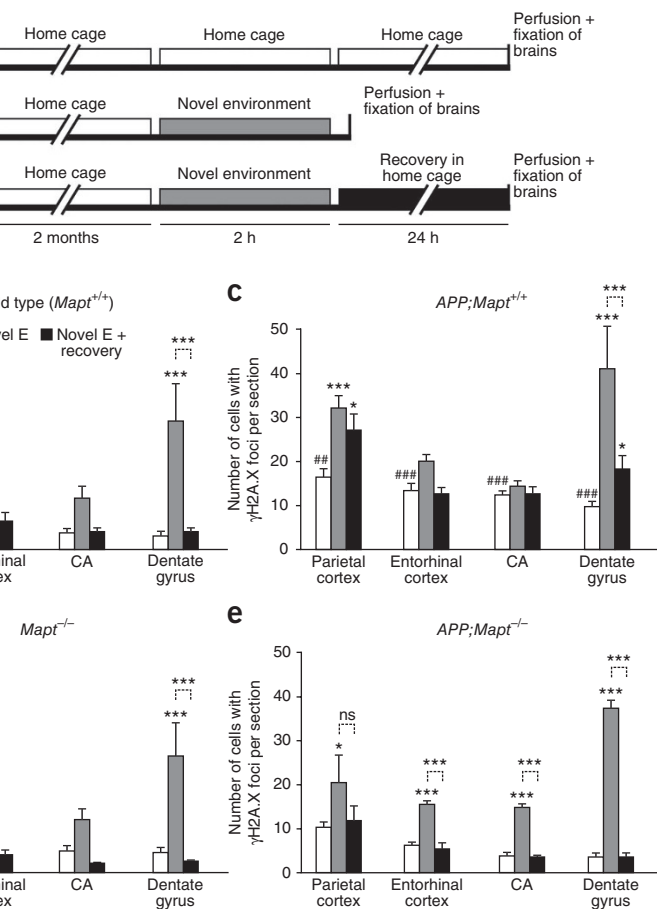
Although γ H2A.X is a well-established marker of DSBs^{4,5,24}, the notion that a natural behavior would increase neuronal DSBs in wild-type mice was perplexing and required further validation. We therefore immunostained brain sections for 53BP1, another widely accepted marker of DSBs^{25,26}. Roughly 85% percent of γ H2A.X foci were positive for 53BP1, and all 53BP1 foci were positive for γ H2A.X (Fig. 1d, Supplementary Fig. 3a–c and data not shown). Because the formation of 53BP1 foci depends on and follows the formation of γ H2A.X foci²⁶, it is possible that the small proportion of γ H2A.X foci that did not co-stain for 53BP1 identifies the most recently formed DSBs. The majority of γ H2A.X+53BP1+ nuclei in the dentate gyrus contained a single focus, and 1% or fewer γ H2A.X+53BP1+ nuclei had three or more foci (Supplementary Fig. 3d,e). These foci were similar in size and shape to neuronal γ H2A.X foci caused by sublethal doses of whole-body γ -irradiation, which we used as a positive control (Supplementary Figs. 3a and 4a).

To further assess whether these foci represent DSBs, we analyzed dentate gyrus homogenates by single-cell gel electrophoresis ('comet assay') (Fig. 3). Because fragmented, but not intact, DNA can be transported

Figure 2 Modulation of DSBs by exploration of a novel environment, overexpression of hAPP/A β , and reduction of endogenous tau. (a) Experimental design (see Online Methods for details). For all experiments, investigators were blinded with respect to the genotype and treatment of mice. E, environment. (b–e) Numbers of γ H2A.X-positive foci in each brain region (averaged from three sections per mouse) in 4- to 6-month-old mice of the indicated genotypes. $n = 4$ –9 mice per genotype and treatment, from two independent experiments. * $P < 0.05$, ** $P < 0.01$ and *** $P < 0.001$ versus mice of the same genotype exposed to the home cage condition, or as indicated by bracket. ## $P < 0.01$ and ### $P < 0.001$ versus wild-type ($Mapt^{+/+}$) mice exposed to the home cage condition. Bonferroni *post hoc* test. Error bars represent means + s.e.m.

by electrophoresis out of the nucleus, nuclei containing fragmented DNA show a comet tail in this assay, whose length reflects the extent of DNA damage (Fig. 3a–c). At alkaline pH, the assay detects both single-strand breaks and DSBs, whereas at neutral pH (as used here), the assay is more specific for DSBs²⁷. The comet assay confirmed that hAPP mice had more cells with DSBs at baseline than wild-type mice, and that exploration of a novel environment increased the number of cells with DSBs in wild-type mice (Fig. 3d). It also allowed us to quantify the extent of DNA damage in greater detail.

After exploration of a novel environment, roughly 30–40% of nuclei had comet tails in both hAPP and wild-type mice (Fig. 3d), but the extent of DNA damage was greater in the hAPP mice, as reflected by longer tails and a greater proportion of nuclear DNA that migrated into the tail (Fig. 3e–g). Independent of genotype and behavioral condition, most comet-bearing nuclei had less than 20% of their DNA in the tail (Fig. 3f) and only 10–15% of them had more than 40% of DNA in the tail (Fig. 3g). This extent of DNA damage was comparable to that induced by the lowest dose of whole-body γ -irradiation used here (Supplementary Fig. 4b,c). Taken together, these findings strongly



support the conclusion that the neuronal γ H2A.X foci we identified in hAPP and wild-type mice represent actual DSBs.

Neuronal activity, but not stress hormones, causes DSBs

To address the mechanisms by which exploration of a novel environment increases neuronal DSBs, we first focused on stress and corticosterone, the latter of which can increase DNA damage *in vitro*²⁸. In comparison with the home cage condition, plasma corticosterone in wild-type mice was elevated after the first 15 min of exploring a novel environment, albeit not as much as after 15 min of restraint

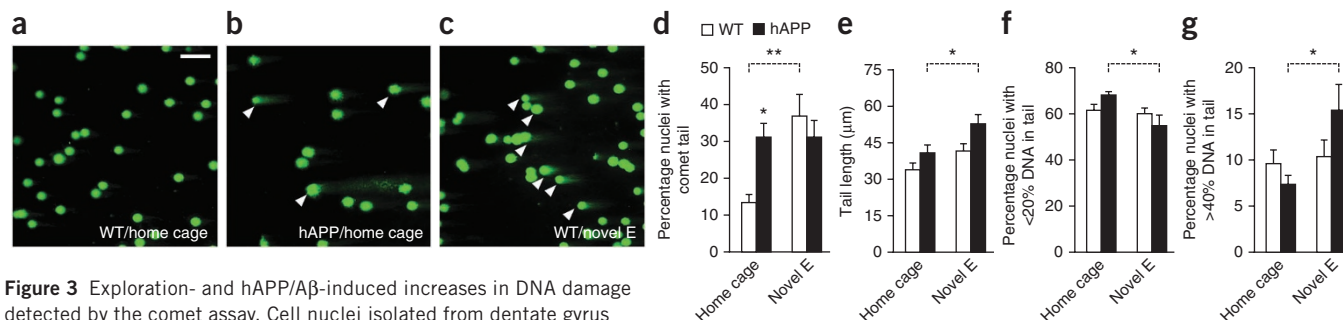
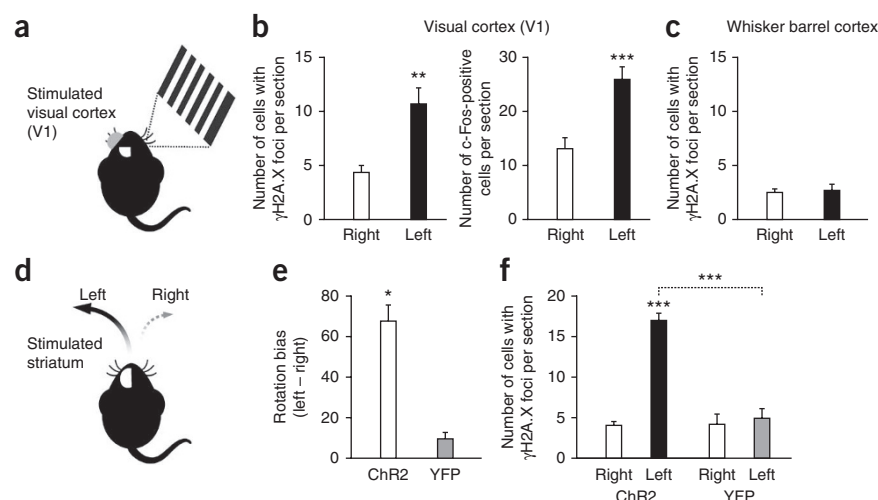


Figure 3 Exploration- and hAPP/A β -induced increases in DNA damage detected by the comet assay. Cell nuclei isolated from dentate gyrus homogenates were assessed for DSB levels by the comet assay at neutral pH. (a–c) Representative images of cell nuclei from wild-type (WT; a) and hAPP (b) mice taken from their home cage (control condition), and from a WT mouse that had just explored a novel environment (novel E) for 2 h (c). DNA of the agarose-embedded nuclei was stained with SYBR-green after separation of DNA fragments by electrophoresis. Images were captured by fluorescence microscopy. Scale bar, 50 μ m. (d) Quantification of the percentage of nuclei with comet tails, which indicate DSBs. For each mouse, a total of 600–800 nuclei in two fields were inspected and scored. (e) Tail length (in μ m), indicating the extent of DNA fragmentation, measured for each cell showing a comet. (f,g) Proportion of comet-bearing nuclei with <20% (f) or >40% (g) fragmented DNA. $n = 6$ –9 mice per genotype and condition. * $P < 0.05$ and ** $P < 0.01$ versus leftmost bar or as indicated by bracket (Bonferroni *post hoc* test); ns, not significant. Error bars represent means + s.e.m.

Figure 4 Network-specific modulation of DSBs by stimulation of primary visual cortex or striatal indirect-pathway neurons. (a–c) Modulation of DSBs by unilateral visual stimulation of 4-month-old anesthetized mice ($n = 6$). (a) Experimental design (see Online Methods for details). Only the right eye was stimulated, whereas the left eye was shielded from light. (b) Numbers of cells with γ H2A.X-positive foci (left) and of c-Fos-positive cells (right) in V1 (averaged from three sections per mouse). (c) Numbers of cells with γ H2A.X-positive foci in the whisker-barrel cortex of the same mice (averaged from three sections per mouse). (d–f) Modulation of DSBs and motor behavior by unilateral optogenetic stimulation of the striatum in awake, behaving mice. (d) Experimental design. Two-month-old mice expressing ChR2-YFP (ChR2) in indirect pathway neurons were compared to age-matched mice expressing YFP alone ($n = 3$ –6 mice per group). The turning behavior of mice was recorded after stimulation of the left striatum. (e) Difference between the numbers of times mice turned left versus right in 1 h. Indirect-pathway stimulation induced ipsilateral rotations. The slight trend of YFP mice to turn more left than right was not significant; one-sample t -test. (f) Numbers of cells with γ H2A.X-positive foci in the right and left striatum (averaged from three sections per mouse). * $P < 0.05$, ** $P < 0.01$ and *** $P < 0.001$ versus leftmost bar by two-tailed paired Student's t -test (b,c), by one-tailed t -test (e) or as indicated by brackets (f; Tukey-Kramer test). Error bars represent means + s.e.m.



stress (Supplementary Fig. 5a). To determine whether the increase in neuronal DSBs after exploration of a novel environment is caused by increases in corticosterone or other factors released from the adrenal gland in response to stress, we analyzed wild-type mice after sham operation or adrenalectomy. The latter group was implanted with subcutaneous corticosterone pellets to hold their corticosterone constant at physiological baseline levels. Sham-operated, but not adrenalectomized, mice showed the expected increases in corticosterone after exploring a novel environment for 2 h (Supplementary Fig. 5b) or being subjected to restraint stress for 15 min (data not shown). However, the two groups of mice showed similar increases in neurons with DSBs in the dentate gyrus and CA regions of the hippocampus after exploring the novel environment, as compared to mice kept in their home cages (Supplementary Fig. 5c,d). Thus, exploration-induced increases in DSBs are not caused by stress-induced surges in corticosterone or other adrenal factors.

We then wondered whether increases in neuronal activity might be sufficient to cause DSBs. To test this hypothesis, we used two

experimental models: activation of the primary visual cortex (V1) by exposure of anesthetized mice to visual stimuli and activation of the striatum in awake, behaving mice by optogenetic stimulation. In the first experiment, one eye was exposed to visual stimuli for 15 min while the other was shielded from light (Fig. 4a). One hour after the visual stimulation began, the number of cells with γ H2A.X foci in the stimulated contralateral V1 was roughly twice that in the unstimulated ipsilateral V1 (Fig. 4b). The number of c-Fos-expressing neurons served as a positive control and increased to a similar extent in the contralateral V1 (Fig. 4b). The whisker barrel cortex served as a stimulus-irrelevant negative control and showed no difference between hemispheres in the number of cells with γ H2A.X foci (Fig. 4c).

To optogenetically modulate neuronal activity in the striatum, transgenic *Adora2a*-Cre mice, which express Cre recombinase in the medium spiny projection neurons of the indirect motor pathway in the dorsomedial striatum, received unilateral striatal injections of adeno-associated virus expressing double-*loxP*-flanked inverted constructs encoding a channelrhodopsin2 (ChR2)-YFP fusion protein or YFP alone. When the transduced striatum was stimulated with light through an implanted optical fiber, ChR2-YFP mice displayed many more ipsiversive rotations than YFP-expressing controls (Fig. 4d,e). Consistent with this behavioral evidence for indirect pathway activation,

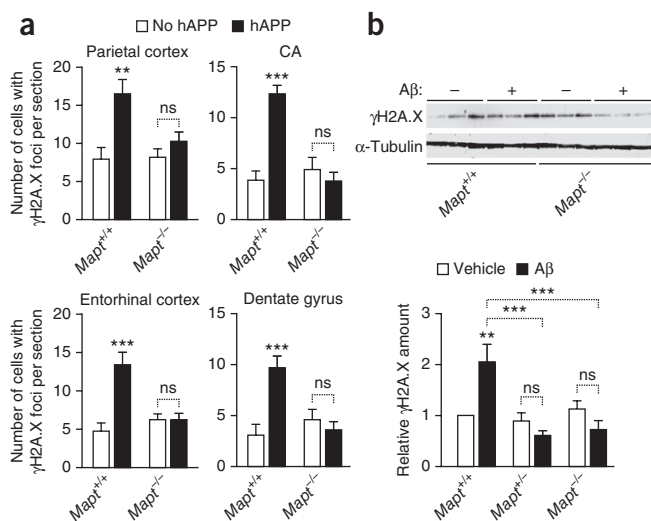


Figure 5 Tau reduction prevents the Aβ-induced increase in neuronal γ H2A.X foci. (a) Numbers of cells with γ H2A.X-positive foci per section (averaged from three sections per mouse) in different brain regions of 4-month-old *Mapt*^{+/+} and *Mapt*^{-/-} mice with or without hAPP expression ($n = 5$ –9 mice per genotype). (b) Western blot analysis of γ H2A.X in primary forebrain neurons cultured for 15 days *in vitro* (DIV 15) from mice of the indicated *Mapt* genotypes, treated with 1 μ M Aβ oligomers (+) or vehicle (–). Representative western blots (top) show results from five culture wells and two genotypes (*Mapt*^{+/+} and *Mapt*^{-/-}). Quantifications of western blot signals (bottom) were from 30 culture wells and three genotypes (*Mapt*^{+/+}, *Mapt*^{-/-} and *Mapt*^{-/-}). $n = 5$ wells per condition from three independent experiments. ** $P < 0.01$ and *** $P < 0.001$ versus leftmost bar or as indicated by brackets (Bonferroni *post hoc* test); ns, not significant. Error bars represent means + s.e.m.

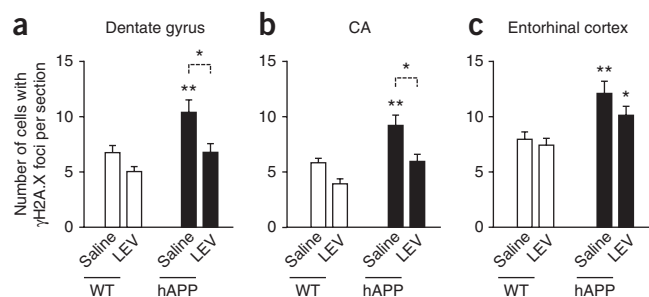


Figure 6 Levetiracetam normalizes the number of γ H2A.X foci in the hippocampus of hAPP mice. (a–c) Numbers of cells with γ H2A.X-positive foci per section (averaged from three sections per mouse) in 4- to 5-month-old hAPP mice ($n = 5$ –8 per treatment) and wild-type controls ($n = 5$ or 6 per treatment) treated for 28 d with saline or levetiracetam (LEV; 75 mg kg^{-1} d^{-1} by subcutaneous Alzet minipump). * $P < 0.05$ and ** $P < 0.01$ versus wild-type or as indicated by brackets (Bonferroni *post hoc* test). Two-way ANOVA revealed effects of treatment and genotype ($P < 0.01$). No significant interaction was observed between them in the hippocampus, and an effect of genotype only ($P = 0.0002$) in entorhinal cortex. Error bars represent means + s.e.m.

illuminated Chr2-YFP-expressing striata contained many more cells with γ H2A.X foci than unilluminated contralateral striata or illuminated striata expressing only YFP (Fig. 4f). Thus, elevating neuronal activity in specific neuronal networks causes an increase in neuronal DSBs selectively in these networks, but not globally.

Preventing aberrant activity in hAPP mice reduces DSBs

hAPP-J20 mice and other lines of hAPP mice show aberrant network activity, including epileptiform spikes and nonconvulsive seizures^{18,29–31}, which may relate to the increased incidence of seizures in humans with Alzheimer's disease³². In hAPP-J20 mice, such aberrant activity can be detected by 6 weeks of age (data not shown), suggesting that it might be causally related to the development of other abnormalities that become manifest around the same time or thereafter. Consistent with this notion, interventions that prevent or suppress this network dysfunction, such as genetic reduction of tau or treatment with the anti-epilepsy drug levetiracetam, improve both synaptic and cognitive functions in hAPP mice^{19,30,33,34}. To determine whether tau reduction prevents A β from causing neuronal DSBs, we crossed hAPP mice onto a tau knockout (*Mapt*^{−/−}) background as described¹⁹ and compared the numbers of neurons with γ H2A.X foci in different brain regions of mice that did or did not express tau and that were exposed to various behavioral conditions (Fig. 2a).

In mice without hAPP expression, tau ablation did not significantly change the number of γ H2A.X-positive neurons at baseline or

after recovery compared to that in wild-type mice (Figs. 2b,d and 5a and Supplementary Fig. 2a,c). In hAPP mice, tau ablation normalized the number of γ H2A.X-positive neurons in all brain regions to values found in wild-type mice, both at baseline and after recovery (Figs. 2e and 5a and Supplementary Fig. 2a,c). In hAPP mice, tau ablation also brought the increase in γ H2A.X-positive neurons in the dentate gyrus immediately after exploration closer to that seen in wild-type controls (Supplementary Fig. 2b). There was a similar trend in hippocampal CA regions (Supplementary Fig. 2b). The presence of A β oligomers increased γ H2A.X in primary neuronal cultures from *Mapt*^{+/+} mice but not in those from *Mapt*^{+/-} or *Mapt*^{−/−} mice (Fig. 5b). Thus, even partial tau reduction prevented A β from increasing DSBs in neurons, consistent with the beneficial effects of partial tau reduction on other A β -dependent abnormalities, including epileptiform activity, premature mortality, cognitive deficits and impairments in axonal transport^{19,30,33}.

As tau reduction prevents aberrant network activity in different lines of hAPP transgenic mice³³, the above findings support the hypothesis that the increased neuronal DSBs in hAPP mice are caused by such activity. To test this hypothesis more directly, we treated 4- to 5-month-old mice with the anti-epilepsy drug levetiracetam, which partially suppresses aberrant network activity, reverses synaptic deficits and ameliorates cognitive deficits in hAPP mice³⁴. One month of continuous subcutaneous infusion of levetiracetam (75 mg per kilogram body weight per day) normalized the numbers of γ H2A.X-positive foci in the dentate gyrus and hippocampus of hAPP mice but had no effect on γ H2A.X-positive foci in wild-type mice (Fig. 6a,b). Treated hAPP mice also showed a trend toward fewer γ H2A.X-positive foci in the entorhinal cortex (Fig. 6c).

Role of extrasynaptic NMDARs in formation of neuronal DSBs

In primary cultures of wild-type neurons, addition of A β oligomers increased the number of γ H2A.X-positive cells and γ H2A.X levels (Figs. 1c and 7). This effect was absent in neurons whose activity was reduced with a sodium channel blocker (TTX) or with antagonists of AMPA-type or NMDA-type glutamate receptors (NBQX (2,3-dihydroxy-6-nitro-7-sulfamoylbenzo[f]quinoxaline-2,3-dione) or AP5 ((2R)-amino-5-phosphonovaleric acid), respectively) (Fig. 7a,b).

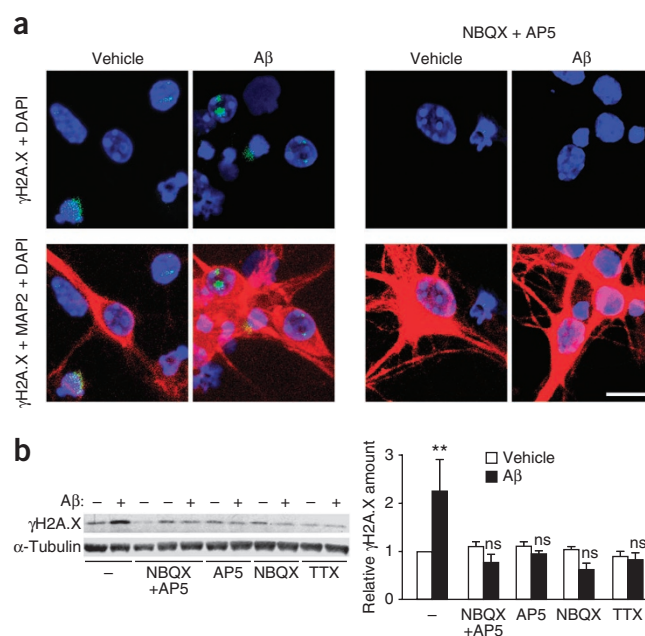
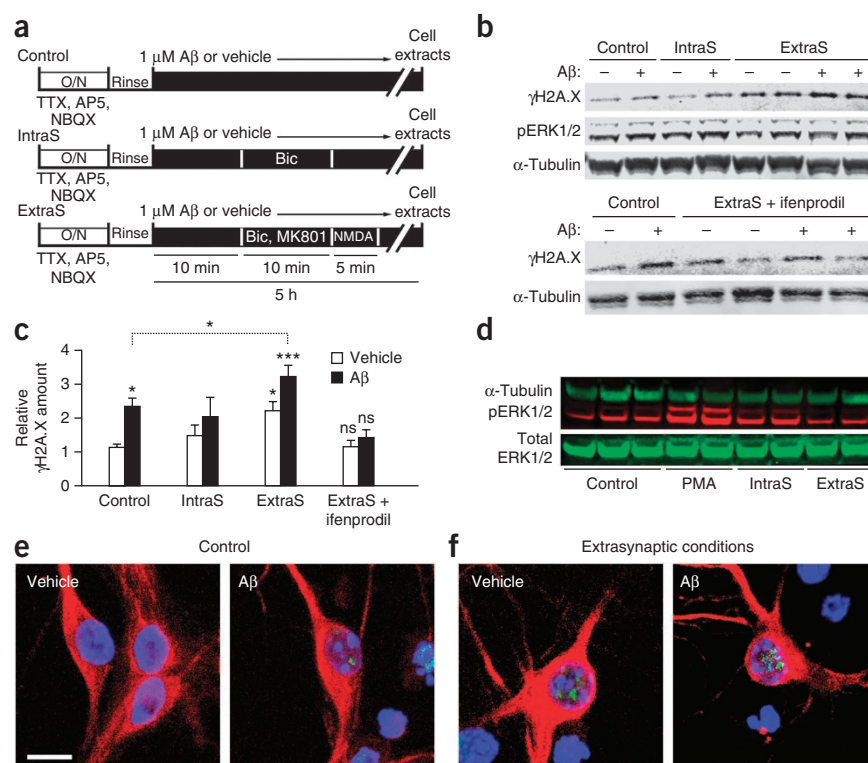


Figure 7 A β -induced increases in γ H2A.X in neuronal cultures depend on neuronal activity. (a) Primary cultures of wild-type mouse hippocampal neurons (day *in vitro* 15), treated with A β oligomers (1 μM) or vehicle for 4 h in the presence or absence of antagonists of AMPARs (NBQX) and NMDARs (AP5) as indicated. Cultures were triple-labeled for γ H2A.X (green), the neuronal marker MAP2 (red) and the nuclear marker DAPI (blue). Scale bar, 10 μm . (b) Western blot analysis of γ H2A.X in primary cultures of wild-type mouse forebrain neurons treated with 1 μM A β oligomers (+) or vehicle (−) for 4 h in the presence or absence (−) of NBQX plus AP5, AP5 alone, NBQX alone, or the sodium channel blocker tetrodotoxin (TTX). Left, representative western blot; right, quantification. $n = 5$ –7 wells per condition from five independent experiments. ** $P < 0.01$ versus leftmost bar (Tukey's *post hoc* test); ns, not significant. Error bars represent means + s.e.m.

Figure 8 A β -induced increases in γ H2A.X-positive foci in primary neuronal cultures require activation of extrasynaptic NR2B-containing NMDARs. (a) Treatment of wild-type mouse hippocampal neuronal cultures (see Online Methods for details). Bic, bicuculline. MK801, dizocilpine. ExtraS, extrasynaptic. IntraS, intrasynaptic. O/N, overnight. (b,c) Western blot analysis of pERK1/2 (b) and γ H2A.X (b,c) in primary neurons treated as indicated. Representative western blots, b; quantification of γ H2A.X, c. $n = 5$ –12 wells per condition from seven independent experiments. * $P < 0.05$ and *** $P < 0.001$ versus

leftmost bar (Dunnett's test) or as indicated by the brackets (two-tailed t -test); ns, not significant. Error bars represent means $\pm$ s.e.m. (d) Representative western blot showing the effects of the different treatments on phosphorylation of ERK1/2 (pERK1/2) 30 min after stimulation of cells in the absence of A β . PMA (12-*O*-tetradecanoylphorbol-13-acetate), a phorbol ester that activates protein kinase C, leading to ERK1/2 phosphorylation, was used as a positive control. As compared with the control condition, pERK1/2 amounts were higher under conditions favoring spontaneous activation of intrasynaptic NMDARs and lower under conditions favoring activation of extrasynaptic NMDARs. (e,f) Confocal microscopy images of immunostained primary cultures of wild-type mouse hippocampal neurons plated at low density, treated with A β oligomers (1 μ M) or vehicle for 4 h under control conditions (e) or conditions favoring the activation of extrasynaptic NMDARs (f), and triple-labeled, for γ H2A.X (green) and DAPI (blue) and for MAP2 (red) to illustrate their neuronal morphology. Scale bar, 10 μ m.



The finding that TTX blocked A β -induced increases in DSBs suggests that this A β effect depends on action potential-induced synaptic release of neurotransmitters. The finding that NBQX blocked the A β effect suggests that postsynaptic depolarization mediated by AMPA receptors is required for this effect. Postsynaptic depolarization is required for removal of the Mg²⁺ block of NMDA receptors, which in turn is required for activation of these receptors. The finding that AP5 blocked the A β effect indicates that it indeed depends on activation of NMDA receptors.

Much evidence suggests that the functional consequences of glutamatergic neurotransmission critically depend on the balance between intra- versus extrasynaptic NMDA receptor activation^{18,35}. To determine whether the A β -induced enhancement of DSBs requires activation of intrasynaptic or extrasynaptic NMDARs, we cultured neurons under conditions that favor the stimulation of one or the other (Fig. 8a). In the absence of A β oligomers, γ H2A.X levels were higher under extrasynaptic-receptor-biased conditions, whereas levels of phosphorylated ERK1/2 were higher under intrasynaptic-receptor-biased conditions (Fig. 8b–d and Supplementary Fig. 6a). Extrasynaptic-receptor-biased conditions were also associated with increased production of reactive oxygen species (ROS), but prevention of this increase by pretreatment of cultures with the ROS scavenger Euk-134 did not diminish the elevation of γ H2A.X (Supplementary Fig. 7).

A β oligomers increased γ H2A.X levels more markedly under extrasynaptic-receptor-biased conditions (Fig. 8b,c), and the increase was prevented by ifenprodil (Fig. 8b,c), a drug that specifically blocks NMDARs containing the NR2B subunit, which primarily reside extrasynaptically³⁶. A β oligomers caused more intense γ H2A.X immunoreactivity in neuronal nuclei under extrasynaptic-receptor-biased than under unbiased control conditions (Fig. 8e,f). Thus, A β -induced

formation of neuronal DSBs required the following (as shown by blockade using the inhibitors indicated in parentheses): action potential-dependent synaptic release of neurotransmitters (TTX), postsynaptic depolarization (NBQX), and activation of NR2B-containing (ifenprodil) NMDARs (AP5). Blocking any of these critical elements disrupted the causal chain that connects A β to neuronal DNA damage and γ H2A.X formation.

DISCUSSION

Our study shows that exploratory activity, which is associated with physiological increases in neuronal activity and the encoding of new information, causes widespread increases in neuronal DSBs. DNA damage in postmitotic cells has previously been demonstrated in pathological contexts^{1,37}. However, to our knowledge, this is the first *in vivo* demonstration of physiologically induced DSBs in postmitotic cells. Four lines of evidence support the presence of DSBs: (i) the formation of distinct nuclear foci immunoreactive for γ H2A.X, a well-established marker of DSBs^{4,5}, (ii) the co-labeling of γ H2A.X foci with 53BP1, another widely accepted marker of DSBs^{25,26}, (iii) the close resemblance of the foci to foci elicited by doses of γ -irradiation known to cause DSBs^{38,39} and (iv) the detection of DNA fragmentation in single cells by comet assay carried out at neutral pH²⁷.

Pathologically elevated A β increased neuronal DSBs at baseline and augmented and prolonged the increases in DSBs that resulted from physiological increases in neuronal activity. It is tempting to speculate that these A β effects could ultimately threaten the stability of the neuronal genome and interfere with the fine regulation of gene expression required for cognitive functions. Consistent with this notion, genetic ablation of a DNA repair enzyme was recently shown to cause memory deficits in mice⁴⁰.

Different approaches used to detect DNA damage and related alterations have different sensitivities. Extensive DNA damage in proapoptotic or apoptotic cells, such as that caused by severe oxidative stress, can be detected by TUNEL staining⁴¹. In contrast, neurons of hAPP mice and wild-type controls were negative by TUNEL, even when they contained γ H2A.X^{53BP1}⁺ foci (data not shown), indicating that the TUNEL method is not sensitive enough to detect the degree or type of DNA damage that occurs in these mice. Previous reports also indicate that γ H2A.X immunostaining is more sensitive than TUNEL at detecting DSBs^{4,11}. In our study, γ H2A.X immunostaining and the comet assay yielded quantitatively similar results. Relative to baseline measures in the dentate gyrus of wild-type mice, hAPP mice had three times more cells with γ H2A.X-immunoreactive foci and 2.5 times more cells with DNA damage by comet assay. These assays provide qualitatively distinct and complementary types of information. For example, the double labeling of brain sections with antibodies to γ H2A.X and NeuN allowed us to determine that most cells with DSBs in wild-type and hAPP mice, before and after exploration of a novel environment, were neurons. After exploration, hAPP mice had numbers of neurons with DSBs similar to those in wild-type controls, as assessed by γ H2A.X immunostaining, but more severe DNA fragmentation per cell, as assessed by comet assay.

Several studies suggest that A β oligomers and Alzheimer's disease can cause neurons to re-enter the cell cycle^{42,43}. However, neuronal cell cycle events in APP transgenic mice have, to our knowledge, not been reported before 6 months of age and are more prominent in the brainstem and frontal cortex than in the dentate gyrus. Furthermore, they appear to be absent in adult wild-type mice. Therefore, it is unlikely that cell cycle re-entry caused the neuronal γ H2A.X^{53BP1}⁺ foci we observed in wild-type and hAPP mice, which were most abundant in the dentate gyrus and detectable in hAPP mice by 1.5 months of age. It is possible, however, that chronic increases in DNA damage ultimately promote cell cycle events.

We found the greatest number of neurons with exploration-induced DSBs in the dentate gyrus, a brain region critical for learning and memory. As exposure to three brief electroconvulsive seizures does not increase the number of DSBs in the dentate gyrus of rats¹², it is remarkable that a natural animal behavior increased DSBs in this region. In all brain regions examined, the DSBs in wild-type mice were transient (lasting <24 h), indicating efficient repair. Therefore, our snapshot measurements likely underestimate the total number of DSBs that occur in neurons over time. We hypothesize that the formation of DSBs in neurons is a natural process that facilitates the extensive chromatin remodeling and changes in gene expression involved in information processing, learning and memory. However, we cannot exclude the possibility that the DSBs are merely a byproduct of energy metabolism in particularly active neurons.

We obtained several lines of evidence in support of the hypothesis that both exploratory behavior and pathologically elevated A β increase neuronal DSBs by modulating neuronal activity: (i) increasing neuronal activity by visual stimulation in the absence of active behaviors or by optogenetic stimulation in behaving mice was sufficient to increase neuronal DSBs, (ii) interventions that suppress aberrant network activity in hAPP mice^{30,34} normalized their levels of neuronal DSBs, and (iii) A β oligomers no longer increased neuronal DSBs in primary cultures when neuronal activity was blocked.

Our findings that tau ablation in hAPP mice prevented both baseline and exploration-associated abnormalities in neuronal DSBs suggest that endogenous tau is essential for their accumulation. Tau ablation also prevents many other problems in this and other lines of hAPP mice, including aberrant network activity, abnormalities

in activity-related proteins, deficits in synaptic functions, impairments in learning and memory, and behavioral alterations^{19,30,33}. Furthermore, in mice without hAPP, tau ablation had very limited effects on neuronal γ H2A.X foci (this study) and on a wide range of other measures^{33,44}, supporting the notion that tau reduction may be of benefit in Alzheimer's disease.

The precise mechanisms by which tau enables A β -induced neuronal dysfunction remain to be determined; however, several lines of evidence suggest that permission or promotion of aberrant excitatory neuronal activity is critical^{17–19,30}. Consistent with this notion, we found that treatment with the anti-epileptic drug levetiracetam also reduced abnormalities in neuronal γ H2A.X foci in hAPP mice.

We used primary cultures to confirm that A β oligomers alone can cause neuronal DSBs and to explore the mechanisms underlying this phenomenon. Several mechanisms have been proposed to explain how A β oligomers may cause aberrant neuronal activity and impair synaptic functions^{18,45}. Most pertinent to the present study, A β oligomers lead to excessive activation of extrasynaptic NMDARs^{36,46}, which could trigger aberrant excitatory neuronal activity as well as synaptic depression^{18,35}, both of which are observed in hAPP mice^{17,45}. In neuronal cultures, we were able to prevent A β oligomer-induced increases in neuronal DSBs by blocking extrasynaptic NR2B-containing NMDARs. These results suggest that activation of NR2B-containing extrasynaptic NMDARs contributes causally to the formation of A β - and neuronal activity-dependent DSBs and that selective blockade of these receptors may help protect and stabilize the neuronal genome in disease-related contexts.

In the absence of A β oligomers, the number of DSBs transiently increased after stimulation of cultured neurons with NMDA, but fell back to baseline within a few hours (**Supplementary Fig. 5b**)¹¹. However, in the presence of A β oligomers, or when intrasynaptic NMDARs were blocked pharmacologically, stimulation with NMDA caused a prolonged elevation in DSBs (**Fig. 8d**). These findings suggest that stimulation of intrasynaptic NMDARs is required for the efficient repair of activity-induced neuronal DSBs and that A β oligomers may interfere with this process. Consistent with these *in vitro* findings, hAPP mice showed an abnormal persistence of exploration-induced DSB in the dentate gyrus and parietal cortex, brain regions that are affected by Alzheimer's disease.

What molecular mechanisms might link NMDAR stimulation to the formation of DSBs in physiological and pathophysiological contexts? Tipping the balance toward extrasynaptic over intrasynaptic NMDAR activity might alter downstream signaling cascades and calcium influx in ways that promote aberrant activation of endonucleases, circumscribed increases in free radical production, or inhibition or failure of the DNA repair machinery. These possibilities are not mutually exclusive and deserve to be further explored. The endonuclease Spo11 generates physiological DSBs during meiosis in the testis²³. The increase in its abundance we observed in the dentate gyrus of wild-type mice that had explored a novel environment suggests that Spo11 may contribute to the generation of activity-induced DSBs in neurons as well.

Oxidative stress can cause DSBs⁵, but hAPP-J20 mice show no evidence of increased oxidative stress in the brain at the ages analyzed here⁴⁷. In rat cortical cultures, transient glutamate-induced neuronal DNA damage was found to be reduced by pretreatment with a mitochondrial superoxide dismutase mimetic⁴⁸, but that study focused on single-strand breaks and base oxidation and did not specifically address neuronal DSBs. Although stimulation of NMDA receptors in the extrasynaptic-receptor-biased condition increased both ROS and γ H2A.X in our neuronal cultures, pretreatment of these cultures with

an ROS scavenger reduced the increase in ROS production but not the increase in γ H2A.X, diminishing the likelihood that ROS are a critical mediator of activity-induced DSBs. However, the methods we used are not sensitive enough to rule out more circumscribed and transient increases in ROS production in specific subcellular compartments.

In light of other evidence that histone and chromatin modifications are important in both cognition and memory disorders^{49,50}, it is tempting to speculate that A β -induced abnormalities in the formation and repair of DSBs similar to those we identified in hAPP mice and neuronal cultures may contribute to Alzheimer's disease-related neurological deficits (**Supplementary Fig. 8**). More research is needed to test this hypothetical framework and to assess the roles of neuronal DSBs in health and disease.

METHODS

Methods and any associated references are available in the [online version of the paper](#).

Note: Supplementary information is available in the [online version of the paper](#).

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AUTHOR CONTRIBUTIONS

E.S. designed and conducted behavioral, immunohistochemical and biochemical analyses. P.E.S. designed and carried out levetiracetam treatments and stress-related studies. A.V.K. designed and conducted optogenetic experiments. X.W. and K.H. provided technical assistance for biochemical analyses. K.E. contributed to statistical analyses. N.D. helped design behavioral experiments. A.C.K. supervised the optogenetic experiments. E.S. and L.M. analyzed and interpreted data and wrote the manuscript. L.M. conceived, supervised and provided funding for the study.

COMPETING FINANCIAL INTERESTS

The authors declare competing financial interests: details accompany the [online version of the paper](#).

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- Brasnjevic, I., Hof, P.R., Steinbusch, H.W. & Schmitz, C. Accumulation of nuclear DNA damage or neuron loss: molecular basis for a new approach to understanding selective neuronal vulnerability in neurodegenerative diseases. *DNA Repair (Amst.)* **7**, 1087–1097 (2008).
- Moreira, P.I. *et al.* Nucleic acid oxidation in Alzheimer disease. *Free Radic. Biol. Med.* **44**, 1493–1505 (2008).
- Lu, T. *et al.* Gene regulation and DNA damage in the ageing human brain. *Nature* **429**, 883–891 (2004).
- Bonner, W.M. *et al.* γ H2AX and cancer. *Nat. Rev. Cancer* **8**, 957–967 (2008).
- Ismail, I.H. & Hendzel, M.J. The γ -H2A.X: is it just a surrogate marker of double-strand breaks or much more? *Environ. Mol. Mutagen.* **49**, 73–82 (2008).
- Kim, D. *et al.* Deregulation of HDAC1 by p25/Cdk5 in neurotoxicity. *Neuron* **60**, 803–817 (2008).
- Fernandez-Capetillo, O. *et al.* H2AX is required for chromatin remodeling and inactivation of sex chromosomes in male mouse meiosis. *Dev. Cell* **4**, 497–508 (2003).
- Lee, S.Y. *et al.* Histone H2AX is required for *Xenopus* anterior neural development: critical role of threonine 16 phosphorylation. *J. Biol. Chem.* **285**, 29525–29534 (2010).
- Yin, B. *et al.* Histone H2AX stabilizes broken DNA strands to suppress chromosome breaks and translocations during V(D)J recombination. *J. Exp. Med.* **206**, 2625–2639 (2009).
- Fernando, R.N. *et al.* Cell cycle restriction by histone H2AX limits proliferation of adult neural stem cells. *Proc. Natl. Acad. Sci. USA* **108**, 5837–5842 (2011).
- Crowe, S.L., Movsesyan, V.A., Jorgensen, T.J. & Kondratyev, A. Rapid phosphorylation of histone H2A.X following ionotropic glutamate receptor activation. *Eur. J. Neurosci.* **23**, 2351–2361 (2006).
- Crowe, S.L., Tsukerman, S., Gale, K., Jorgensen, T.J. & Kondratyev, A.D. Phosphorylation of histone H2A.X as an early marker of neuronal endangerment following seizures in the adult rat brain. *J. Neurosci.* **31**, 7648–7656 (2011).
- Palop, J.J. *et al.* Vulnerability of dentate granule cells to disruption of Arc expression in human amyloid precursor protein transgenic mice. *J. Neurosci.* **25**, 9686–9693 (2005).
- Vazdarjanova, A. *et al.* Spatial exploration induces ARC, a plasticity-related immediate-early gene, only in calcium/calmodulin-dependent protein kinase II-positive principal excitatory and inhibitory neurons of the rat forebrain. *J. Comp. Neurol.* **498**, 317–329 (2006).
- Cheng, I.H. *et al.* Accelerating amyloid- β fibrillization reduces oligomer levels and functional deficits in Alzheimer disease mouse models. *J. Biol. Chem.* **282**, 23818–23828 (2007).
- Palop, J.J. *et al.* Neuronal depletion of calcium-dependent proteins in the dentate gyrus is tightly linked to Alzheimer's disease-related cognitive deficits. *Proc. Natl. Acad. Sci. USA* **100**, 9572–9577 (2003).
- Palop, J.J. *et al.* Aberrant excitatory neuronal activity and compensatory remodeling of inhibitory hippocampal circuits in mouse models of Alzheimer's disease. *Neuron* **55**, 697–711 (2007).
- Palop, J.J. & Mucke, L. Amyloid-beta-induced neuronal dysfunction in Alzheimer's disease: from synapses toward neural networks. *Nat. Neurosci.* **13**, 812–818 (2010).
- Roberson, E.D. *et al.* Reducing endogenous tau ameliorates amyloid β -induced deficits in an Alzheimer's disease mouse model. *Science* **316**, 750–754 (2007).
- Götz, J. & Ittner, L.M. Animal models of Alzheimer's disease and frontotemporal dementia. *Nat. Rev. Neurosci.* **9**, 532–544 (2008).
- Kobayashi, D.T. & Chen, K.S. Behavioral phenotypes of amyloid-based genetically modified mouse models of Alzheimer's disease. *Genes Brain Behav.* **4**, 173–196 (2005).
- Marchetti, C. & Marie, H. Hippocampal synaptic plasticity in Alzheimer's disease: what have we learned so far from transgenic models? *Rev. Neurosci.* **22**, 373–402 (2011).
- Lange, J. *et al.* ATM controls meiotic double-strand-break formation. *Nature* **479**, 237–240 (2011).
- Fillingham, J., Keogh, M.C. & Krogan, N.J. γ H2AX and its role in DNA double-strand break repair. *Biochem. Cell Biol.* **84**, 568–577 (2006).
- Anderson, L., Henderson, C. & Adachi, Y. Phosphorylation and rapid relocalization of 53BP1 to nuclear foci upon DNA damage. *Mol. Cell Biol.* **21**, 1719–1729 (2001).
- Ward, I.M., Minn, K., Jorda, K.G. & Chen, J. Accumulation of checkpoint protein 53BP1 at DNA breaks involves its binding to phosphorylated histone H2AX. *J. Biol. Chem.* **278**, 19579–19582 (2003).
- Olive, P.L. & Banath, J.P. The comet assay: a method to measure DNA damage in individual cells. *Nat. Protoc.* **1**, 23–29 (2006).
- Flint, M.S., Baum, A., Chambers, W.H. & Jenkins, F.J. Induction of DNA damage, alteration of DNA repair and transcriptional activation by stress hormones. *Psychoneuroendocrinology* **32**, 470–479 (2007).
- Harris, J.A. *et al.* Transsynaptic progression of amyloid- β -induced neuronal dysfunction within the entorhinal-hippocampal network. *Neuron* **68**, 428–441 (2010).
- Roberson, E.D. *et al.* Amyloid- β /Fyn-induced synaptic, network, and cognitive impairments depend on Tau levels in multiple mouse models of Alzheimer's disease. *J. Neurosci.* **31**, 700–711 (2011).
- Verret, L. *et al.* Inhibitory interneuron deficit links altered network activity and cognitive dysfunction in Alzheimer model. *Cell* **149**, 708–721 (2012).
- Palop, J.J. & Mucke, L. Epilepsy and cognitive impairments in Alzheimer disease. *Arch. Neurol.* **66**, 435–440 (2009).
- Morris, M., Maeda, S., Vossell, K. & Mucke, L. The many faces of tau. *Neuron* **70**, 410–426 (2011).
- Sanchez, P.E. *et al.* Levetiracetam suppresses neuronal network dysfunction and reverses synaptic and cognitive deficits in an Alzheimer's disease model. *Proc. Natl. Acad. Sci. USA* **109**, E2895–E2903 (2012).
- Hardingham, G.E. & Bading, H. Synaptic versus extrasynaptic NMDA receptor signalling: implications for neurodegenerative disorders. *Nat. Rev. Neurosci.* **11**, 682–696 (2010).
- Li, S. *et al.* Soluble A β oligomers inhibit long-term potentiation through a mechanism involving excessive activation of extrasynaptic NR2B-containing NMDA receptors. *J. Neurosci.* **31**, 6627–6638 (2011).
- Shull, E.R. *et al.* Differential DNA damage signaling accounts for distinct neural apoptotic responses in ATLD and NBS. *Genes Dev.* **23**, 171–180 (2009).
- Bhogal, N., Jalali, F. & Bristow, R.G. Microscopic imaging of DNA repair foci in irradiated normal tissues. *Int. J. Radiat. Biol.* **85**, 732–746 (2009).
- Grudzenski, S., Rath, A., Conrad, S., Rube, C.E. & Lobrich, M. Inducible response required for repair of low-dose radiation damage in human fibroblasts. *Proc. Natl. Acad. Sci. USA* **107**, 14205–14210 (2010).
- Canugovi, C. *et al.* Endonuclease VIII-like 1 (NEIL1) promotes short-term spatial memory retention and protects from ischemic stroke-induced brain dysfunction and death in mice. *Proc. Natl. Acad. Sci. USA* **109**, 14948–14953 (2012).

41. Zhang, Y., Goodyer, C. & LeBlanc, A. Selective and protracted apoptosis in human primary neurons microinjected with active caspase-3, -6, -7, and -8. *J. Neurosci.* **20**, 8384–8389 (2000).
42. Lee, H.G. *et al.* Cell cycle re-entry mediated neurodegeneration and its treatment role in the pathogenesis of Alzheimer's disease. *Neurochem. Int.* **54**, 84–88 (2009).
43. Li, L., Cheung, T., Chen, J. & Herrup, K. A comparative study of five mouse models of Alzheimer's disease: cell cycle events reveal new insights into neurons at risk for death. *Int. J. Alzheimers Dis.* **2011**, 171464 (2011).
44. Morris, M. *et al.* Age-appropriate cognition and subtle dopamine-independent motor deficits in aged Tau knockout mice. *Neurobiol. Aging* doi:10.1016/j.neurobiolaging.2012.12.003 (16 January 2013).
45. Cissé, M. *et al.* Reversing EphB2 depletion rescues cognitive functions in Alzheimer model. *Nature* **469**, 47–52 (2011).
46. Li, S. *et al.* Soluble oligomers of amyloid β -protein facilitate hippocampal long-term depression by disrupting neuronal glutamate uptake. *Neuron* **62**, 788–801 (2009).
47. Esposito, L. *et al.* Reduction in mitochondrial superoxide dismutase modulates Alzheimer's disease-like pathology and accelerates the onset of behavioral changes in human amyloid precursor protein transgenic mice. *J. Neurosci.* **26**, 5167–5179 (2006).
48. Yang, J.L., Tadokoro, T., Keijzers, G., Mattson, M.P. & Bohr, V.A. Neurons efficiently repair glutamate-induced oxidative DNA damage by a process involving CREB-mediated up-regulation of apurinic endonuclease 1. *J. Biol. Chem.* **285**, 28191–28199 (2010).
49. Day, J.J. & Sweatt, J.D. Epigenetic mechanisms in cognition. *Neuron* **70**, 813–829 (2011).
50. Gräff, J., Kim, D., Dobbin, M.M. & Tsai, L.H. Epigenetic regulation of gene expression in physiological and pathological brain processes. *Physiol. Rev.* **91**, 603–649 (2011).

ONLINE METHODS

Mice and tissues. We studied 4–7-month-old heterozygous hAPP transgenic and wild-type mice from line J20 on a *Mapt*^{+/+}, *Mapt*^{+/-} or *Mapt*^{-/-} C57Bl/6J background. Transgenic C57Bl/6J mice expressing Cre recombinase in medium spiny neurons of the indirect pathway in the dorsomedial striatum directed by *Adora2a* (A2a) receptor regulatory elements (A2A-Cre mice) were from GENSAT. Eight-week-old adrenalectomized versus sham-operated wild-type mice were obtained from the Jackson Laboratory. All *in vivo* analyses were carried out on sex-matched or sex-balanced groups. Mice were anesthetized with Avertin (tribromoethanol, 250 mg/kg) and perfused transcardially with 0.9% NaCl. One hemibrain was snap-frozen and stored at -80 °C. The other was drop-fixed in 4% paraformaldehyde in phosphate-buffered saline (PBS) and sectioned into 30-μm-thick slices using a sliding microtome (Leica SM2000R). After visual and optogenetic stimulation, both hemibrains were fixed with paraformaldehyde. A total of 205 mice were used in this study. All animal experiments were approved by the Institutional Animal Care and Use Committee of the University of California, San Francisco.

Antibodies and other reagents. Primary antibodies used included monoclonal mouse anti-γH2A.X (JBW301, Millipore), anti-NeuN biotinylated (MAB377B, Millipore) and anti-α-tubulin (B5-1-2, Sigma-Aldrich), and polyclonal rabbit anti-MAP2 (AB5622, Millipore), anti-phospho-ERK1/2 (9101, Cell Signaling Technology) and anti-total ERK1/2 (4696, Cell Signaling Technology), anti-53BP1 (NB100304, Novus Biologicals) and anti-Spo11 (NBPI-58172, Novus Biologicals). Synthetic Aβ₁₋₄₂ oligomers were prepared from lyophilized monomers (rPeptide or Biopetide), and naturally secreted Aβ oligomers were isolated from stably hAPP-transfected CHO-7PA2 cells⁴⁵. Tetrodotoxin (TTX), bicuculline, NMDA and the following inhibitors of glutamate receptors were from Tocris: 2,3-dihydroxy-6-nitro-7-sulfamoylbenzo[f]quinoxaline-2,3-dione (NBQX), (2R)-amino-5-phosphonovaleric acid (AP5), dizocilpine (MK801), and ifenprodil. Levetiracetam (UCB Pharmaceuticals) was infused continuously with subcutaneous Alzet minipumps³⁴.

Cell cultures and treatments. Primary cultures of hippocampal and cortical neurons established from postnatal day 0 pups^{45,51} were plated on dishes (BD Falcon) (10⁶ cells per 35-mm dish) or glass coverslips (Bellco Glass) (150,000 cells per 12-mm coverslip) coated with poly-L-lysine (Sigma-Aldrich) and laminin (Roche). Cultures were used for experiments after 14 days *in vitro* (DIV) and incubated at 37 °C during all treatments. They were incubated overnight with NBQX (50 μM) and D-AP5 (100 μM). Cells were then rinsed with warm PBS and incubated for 60 min at 37 °C in neuron-conditioned medium that did or did not contain TTX (1 μM), or NBQX (50 μM) and/or D-AP5 (100 μM), followed by addition of Aβ oligomers (1 μM) or vehicle. After 5 h, cells were processed for western blot analysis or immunofluorescence staining.

To examine the roles of intrasynaptic versus extrasynaptic NMDA receptors, cells were incubated with TTX (1 μM), NBQX (50 μM) and D-AP5 (100 μM) for 24 h. They were then washed in warm PBS and incubated in neuron-conditioned medium for 30–60 min to remove the inhibitors and prevent them from interfering with stimulation of glutamate receptors. Aβ oligomers (1 μM) or vehicle was added to the cultures 10 min before stimulation. At this time, half of the medium was collected from each well and kept at 37 °C (conditioned medium).

To activate synaptic NMDARs ('intrasynaptic' or IntraS protocol), neurons were disinhibited with 10 μM bicuculline for 10 min, rinsed with warm PBS and incubated in their conditioned medium. To activate extrasynaptic NMDARs ('extrasynaptic' or ExtraS protocol), neurons were incubated in medium containing MK801 (40 μM) and bicuculline (10 μM) for 10 min, rinsed with warm PBS, incubated in medium containing NMDA (10 μM) for 5 min, washed and then incubated in their conditioned medium. Five hours later, cells were processed for western blot analysis or immunofluorescence staining. For specific inhibition of NR2B-containing NMDARs, the extrasynaptic protocol was modified by adding ifenprodil (3 μM) to all steps, starting with the overnight inhibitor pretreatment.

Western blot analysis. Frozen hemibrains from perfused mice were microdissected in ice-cold PBS containing complete protease inhibitors (Roche) and homogenized in RIPA buffer (10 mM Tris-HCl, pH 7.5, 50 mM NaCl, 30 mM Na₂P₂O₇, 5 μM ZnSO₄, 10% (vol/vol) glycerol, 0.1% SDS, 1% (vol/vol) Triton X-100, 1 mM dithiothreitol) containing complete protease inhibitors and

a cocktail of phosphatase inhibitors (Sigma-Aldrich). Cultured cells were washed in PBS and scraped into RIPA buffer. Lysates were sonicated for 5 min at 4 °C and spun in a refrigerated microcentrifuge at maximum speed for 10 min. Protein concentrations were determined by Bradford assay. Proteins (20 μg per sample) were loaded on 4–12% Bis-Tris gels (Invitrogen), separated by SDS-PAGE and transferred to nitrocellulose membranes. After 1 h in blocking solution (5% (wt/vol) nonfat dry milk in TBS), membranes were incubated with primary antibodies (γH2A.X, total ERK1/2 and Spo11 diluted 1:1,000; pERK1/2, 1:500; α-tubulin, 1:50,000) in TBS containing 0.1% (vol/vol) Tween-20 (TBS-T) and 5% nonfat milk. Blots were washed three times in TBS-T and incubated with secondary IRD-tagged antibodies (800 IRDye and 680 IRDye; Li-COR, Lincoln, NE, 1:5,000) in Odyssey blocking buffer for 1 h at room temperature (20–22 °C; RT). Western blot signals were analyzed with an Odyssey Li-COR laser scanning and imaging system (Li-COR, v3.0).

Behavioral conditions and neuronal stimulation. *Exploration of a novel environment.* Mice were housed in groups of four per cage. Mice in the control group were kept in their original home cages. Mice in the novel environment (Novel E) and the Novel E + recovery groups were moved to the testing room and allowed to acclimate to the room for at least 1 h. Each set of four group-housed mice was then transferred together to a new cage, which was larger and contained different litter, odors, stations and toys, and allowed to explore the cage for 2 h. Introducing mice into a novel environment alongside other mice that they know well decreases their anxiety and increases their exploratory activity. After the novel environment exploration, mice in the Novel E group were killed, whereas mice in the Novel E + recovery group were transferred back into their home cages and kept there for 24 h before being killed. Half of the control mice were killed together with the Novel E group and the other half with the Novel E + recovery group.

Restraint stress. Each mouse was placed into a 3.0 × 11.5 cm cylinder with porous walls. Pads were added to further restrict space. After 15 min of restraint, the cylinder was placed in an isoflurane chamber. Anesthetized mice were then removed, injected with Avertin and perfused transcardially within 10 min after the restraint stress ended.

Visual stimulation. Mice were anesthetized with 5% isoflurane in a sealed box and placed in a stereotaxic frame in which anesthesia was maintained with isoflurane (0.8%) using a mask. Transparent gel was put on both eyes and one eye was shielded from light with a cap. The other eye was kept open with mini-tweezers. Care was taken not to touch the whiskers to avoid stimulation of the barrel field cortex. Visual stimulation consisted of different patterns of horizontal or vertical black-and-white gratings generated by a PowerPoint animation on a computer screen. The screen was positioned 25 cm from the open eye perpendicular to its visual axis. Each pattern was presented three times for 0.5 s with 0.2 s of darkness between presentations to avoid persistent light exposure. After 15 min of visual stimulation, the screen was shut off and mice were kept in the stereotaxic frame for another 15 min. They were then injected with Avertin, removed from isoflurane and perfused transcardially 30 min later. All procedures were performed under minimal ambient lighting.

Optogenetic stimulation of the striatum. Optogenetic stimulation was carried out as described⁵². Briefly, A2A-Cre mice received unilateral stereotaxic striatal injections with adeno-associated virus expressing double-floxed inverted constructs encoding a ChR2-YFP fusion protein or YFP alone. A glass fiber was then implanted close to the viral injection site. Two weeks later, awake mice were placed in a 16 × 16 inch square box, connected to a 473-nm laser via a 1-m optical fiber and stimulated for 1 h with a pattern of light (1 mW) 30 s constant on and 30 s off. All mice were habituated to the box and fiber tethering for 3 d (30–60 min per day) before the laser stimulation. Video tracking of mice during the stimulation sessions and analysis were performed with Ethovision 7.0 software⁵³ to monitor rotational behaviors. One hour after the end of the stimulation protocol, mice were anesthetized with ketamine and xylazine and transcardially perfused with PBS followed by 4% paraformaldehyde. The brains were then processed for immunohistochemistry.

Immunofluorescence staining. Primary neurons were rinsed with PBS, fixed in 100% methanol for 10 min and rinsed in Tris-buffered saline (TBS). Coverslips

were incubated in blocking solution (5% normal donkey serum in 0.1% TBS-Tween) for 60 min at RT and then overnight at 4 °C with antibodies to γ H2A.X and MAP2 diluted at 1:400 in blocking solution. After rinses with 0.1% TBS-Tween, cells were incubated with corresponding Alexa-conjugated secondary antibodies (1:300, A22202 and A21207, Invitrogen) diluted in 4% normal donkey serum in TBS for 1 h at RT.

Preparation and immunohistochemistry of brain sections were performed essentially as described²⁹. However, PBS was replaced with TBS, and steps were added for antigen retrieval and peroxidase inhibition to enhance nuclear staining. Sections were incubated overnight at 4 °C with primary antibodies diluted at 1:400 (anti- γ H2A.X alone or combined with anti-53BP1 or anti-NeuN-biotin). Secondary antibodies were donkey anti-mouse Alexa 488 (A22202, Invitrogen) and anti-rabbit Alexa 593 (A21207, Invitrogen, 1:500) and were used with streptavidin-Cy3 (Invitrogen). After extensive rinses, coverslips and sections were mounted with Vectashield mounting medium with DAPI (Vector Laboratories).

Colocalization of γ H2A.X staining and DAPI allowed us to selectively analyze nuclei. Cells containing at least one γ H2A.X-positive focus in the nucleus were counted in four brain regions. For each mouse and brain region, the average number of cells with foci was calculated from counts obtained in three coronal sections. For the optogenetic stimulation experiment, the average number of cells with γ H2A.X-positive foci was calculated from counts obtained in three striatal fields per mouse. Each field was located in a different coronal section and covered an area of viral transduction as judged by the presence of YFP fluorescence.

Digitized images were obtained with an Axiocam HRc camera (Zeiss) mounted on a BX-60 microscope (Olympus) using Axiovision LE software. Confocal images were acquired with an MC04 Zeiss LSM510 multiphoton microscope with Meta software. Stacks of confocal images were deconvolved with Huygens software. The z-maximum-intensity projection function of ImageJ was used to optimize the appearance of DAPI labels. To visualize γ H2A.X or 53BP1 foci, we selected micrographs in the center of the z-stacks.

Single-cell gel electrophoresis (comet assay). After perfusion of mice with cold saline, brains were removed rapidly and briefly stored in ice-cold Hibernate A medium (Brainbits) until micro-dissection. Dentate gyri were isolated in ice-cold Hibernate A and homogenized in 1 ml of ice-cold PBS (lacking Ca^{2+} and Mg^{2+} to inhibit nucleases) by 30 strokes of pipetting. For each sample, 25 μ l of homogenate were combined with 250 μ l of low-melting point molten agarose (Trevigen), and 50 μ l were applied per well on comet slides (Trevigen). All steps were performed within an hour of perfusion. Cell lysis and electrophoretic separation of DNA fragments released from cells were performed according to the instructions included in the CometAssay kit (Trevigen). Comet assays were performed at pH 7.4 because this condition makes the assay more specific for the detection of DSBs²⁷. The pH was measured in the TBE buffer used for equilibration and in the TBE buffer of the electrophoresis tank at the beginning and end of the electrophoresis, and was found to be stable in the 7.4–8.0 range. Single-cell gel electrophoresis was performed at 1 V/cm for 40 min. SYBR-green was used to fluorescently stain the comet-like smears of DNA fragments that emerged from cells containing DSBs. Digitized images were obtained with an Axiocam HRc camera (Zeiss) mounted on a BX-60 microscope (Olympus) using Axiovision LE software and analyzed with CometScore software (Tritek). For each sample, the number of cells with comets was counted manually in 2–3 fields and normalized to the total number of cells in the fields. For each field, at least 150 cells were inspected. For each comet detected, the comet was framed, and a cursor adjusted on the head of the comet using the selection tool of the software. The software automatically determined the tail length and percentage of DNA in the tail using a built-in algorithm for comet scoring. For each field, the experimenter calculated the average value of the tail length, sorted the proportion of DNA in the tail in three categories (<20%, 20% to <40%, $\geq$ 40%) and calculated the proportion of comet-bearing cells in each of these categories.

Whole body γ -irradiation. Two to four mice were positioned evenly in a container with individual casings and placed on the rotating support of a Mark I,

series-30 γ -irradiator with a 5,500-Ci cesium-137 source (J.S. Shepherd and Associates). They received 0, 1, 2, 5 or 10 Gy of whole body γ -irradiation. All mice spent the same amount of time in the irradiator, as doses were determined by the progressive removal of attenuators ($\times 2$, $\times 5$, $\times 10$). After irradiation, mice were returned to their home cages for 1 h before anesthesia and transcardial perfusion. One hemisphere was processed for neutral pH comet assay and the other for immunohistochemistry.

Measurement of plasma corticosterone levels by ELISA. Mice were anesthetized with isoflurane (5% in a sealed cage) and injected with Avertin to maintain anesthesia during terminal transcardial perfusions. Blood was collected centrally by puncture of the heart into heparin-coated tubes and centrifuged at 2,000 rpm for 8 min at 4 °C. Plasma was collected and frozen at –80 °C until measurement of corticosterone with an ELISA kit (ENZO Life Sciences) according to the manufacturer's instructions. Mice were perfused with 0.9% saline, and their brains processed for immunohistochemistry.

Assessment of ROS production by live fluorescence microscopy. Neurons grown on coverslips (14 DIV) were pretreated for 1 h with 20 μ M of the ROS scavenger Euk-134 (refs. 54,55; Cayman Chemicals) or vehicle (DMSO), followed by the treatments indicated in **Figure 8a**. H_2O_2 (100 μ M) was used as a positive control for ROS induction. One-and-one-half hours after the end of these treatments, cultures were incubated with 2 μ M of the ROS indicator CM-H2CFDA (Invitrogen) for 40 min, with addition of 4 μ M Hoechst 33342 (Thermo Scientific) during the last 5 min, and then rinsed twice with warm Tyrode buffer (126.5 mM NaCl, 2.5 mM KCl, 2 mM CaCl_2 , 2 mM MgCl_2 , 10 mM HEPES, 30 mM glucose, pH 7.4). A fluorescence microscope (Nikon Eclipse Ti) equipped with a chamber maintained at 37 °C was used to image the live neurons. Images were acquired with Metamorph software (version 7.7.3.0). The plug-in module for particle counting from Image J was used to automatically count the number of Hoechst 33342-labeled nuclei. Nuclei colabeled with CM-H2CFDA were counted manually.

Blind coding and statistical analysis. Investigators who obtained data were blinded with respect to the genotype and treatment of mice and cell cultures. Sample sizes were chosen on the basis of pilot experiments and previous experience with similar types of experiments. Statistical analyses were performed with GraphPad Prism, SPSS or R (R Development Core Team, 2011). Normal distribution of the data was verified by D'Agostino and Pearson omnibus normality test. Differences between two means were assessed by unpaired or paired Student's *t*-test. Differences among multiple means were assessed, as indicated, by one-way or two-way ANOVA, followed by Tukey's, Dunnett's, Bonferroni's or Dunn's *post hoc* tests. For the statistical analysis in **Supplementary Figure 2**, factors were assessed by a four-way repeated-measures ANOVA and a corresponding linear mixed model (package nlme)⁵⁶ that accounted for the effect of repeated measures obtained in the brains of the same mice. Null hypotheses were rejected at the 0.05 level.

51. Cheng, J.S. *et al.* Collagen VI protects neurons against A β toxicity. *Nat. Neurosci.* **12**, 119–121 (2009).
52. Kravitz, A.V. *et al.* Regulation of parkinsonian motor behaviours by optogenetic control of basal ganglia circuitry. *Nature* **466**, 622–626 (2010).
53. Spink, A.J., Tegelenbosch, R.A.J., Buma, M.O.S. & Noldus, L.P.J.J. The EthoVision video tracking system—a tool for behavioral phenotyping of transgenic mice. *Physiol. Behav.* **73**, 731–744 (2001).
54. Pong, K., Doctrow, S.R., Huffman, K., Adinolfi, C.A. & Baudry, M. Attenuation of staurosporine-induced apoptosis, oxidative stress, and mitochondrial dysfunction by synthetic superoxide dismutase and catalase mimetics, in cultured cortical neurons. *Exp. Neurol.* **171**, 84–97 (2001).
55. Rong, Y., Doctrow, S.R., Tocco, G. & Baudry, M. EUK-134, a synthetic superoxide dismutase and catalase mimetic, prevents oxidative stress and attenuates kainate-induced neuropathology. *Proc. Natl. Acad. Sci. USA* **96**, 9897–9902 (1999).
56. Pinheiro, J., Bates, D., DebRoy, S., Sarkar, D. & R Development Core Team. nlme: linear and nonlinear mixed effects models. R package version 3.1-102 (2011).